

Measuring community significance and state vulnerability after nuclear winter

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Abstract

Ensuring the stability of global food trade networks is essential to minimize loss of life after any disaster of international scale. Particularly for nuclear war, whose soot injections would bring long-term climatic consequences and agricultural failure, modeling likely post-catastrophe trade networks and associated vulnerabilities can help policymakers strengthen domestic food security through fostering adequate preparedness. My work first combines a food trade network model with a robust community detection test to critique the statistical significance of trade communities generated by the Louvain method before and after 5 Tg and 47 Tg nuclear catastrophes. This is done separately for wheat, rice, soybeans (also known as soya beans), and maize: four of the largest global calorie sources. Between the weights of 5 Tg and 47 Tg, I find an increase in the percentage of significant trade communities as well as broadly that soy and especially rice communities are most consistently insignificant. I show also that pre-disaster base communities tend to be less significant, aligning with the potential for nuclear winter to strongly reduce food exchange diversity. With these results and other factors, I build a comprehensive vulnerability score function that ranks states at greatest risk to food insecurity. This function places northern European and central Asian states as most vulnerable with interesting variation in geography and severity between catastrophe weights that I motivate by looking individually at driving factors for each country's standardized score. Ultimately, I recommend that international policymakers, especially those from states with high vulnerability scores, use this analysis to identify, personalize, and improve gaps in post-disaster food security such that lives can be saved if such a disaster ever occurs.

1 Introduction

Global food security is immutably linked to climate change by agriculture, whose spatial distribution reflects a dependence on appropriate growing conditions for strong yields. Because of the length of the growing season, quick redistribution and planting after extreme climatic events is unfeasible. This makes the relationship between climate change and starvation of paramount concern. Much work exists that outlines the consequences of anthropogenic and nuclear climate change on state-specific agricultural yields, but most assume a zero export post-disaster policy. I briefly outline these sources as well as previous literature on state food vulnerabilities and network community detection methods to best understand my contribution to the field.

1.1 Spatial impacts of anthropogenic climate change

Existing literature has taken advantage of the wealth of historical domestic and global gridded climate and yield data over the past half-century to link yield changes at the subnational level to various consequences of anthropogenic climate change. Early in the literature, for example, [Schlenker and Roberts \(2009\)](#) investigate what they find to be a nonlinear relationship between temperature and corn, soybean, and cotton yield by first breaking the United States into Hadley III grid zones east of the 100° meridian and then fitting multiple trained regression models (that employ step functions, 8th order Chebyshev polynomials, and piecewise linear functions respectively as a part of a more complex integral model) against a baseline with no weather covariates to determine the improvement in out-of-sample root mean-squared error (RMSE). This comparison process is similarly performed in later literature that looks between ML algorithms ([Heino et al.](#),

2023, Vogel et al., 2019) like Random Forest or XG-Boost instead. In this paper, though, the 100° meridian is chosen to avoid irrigation as a confounding variable, which later would be found through studying more irrigated regions to help reduce (Heino et al., 2023, Ray et al., 2019, Vogel et al., 2019) the yield impacts of particularly extreme temperature conditions. Additional quadratic covariates for precipitation and technological change are included and partially reflect other similar inputs in models such as the 15-parameter equation devised by Ray et al. (2019). After fitting each model, it was found that, depending on the IPCC emissions scenario (between B1 and A1F1), harvest-area weighted yields (Heino et al., 2023, Ray et al., 2019, Schlenker and Roberts, 2009, Vogel et al., 2019) could decrease between 30% and 82%, yields first increasing gradually with greater exposure to high temperature days before sharply declining after critical points around 30°C.

Broadly speaking, these authors and others have used historical data to build, clean, and test models that then are applied to forecast yield impacts to global climate bins (Heino et al., 2023, Jägermeyr et al., 2021, Ray et al., 2019, Schlenker and Roberts, 2009, Vogel et al., 2019). Time series for growing season mean temperature, mean precipitation, soil moisture (Heino et al., 2023), hot/cold day frequency (Heino et al., 2023, Vogel et al., 2019), min/max temperature, and diurnal temperature range (Vogel et al., 2019), among other covariates, are used to construct mathematical (Ray et al., 2019, Schlenker and Roberts, 2009), ML (Heino et al., 2023, Vogel et al., 2019), or simulation models that project gridded yield losses at a global scale. The time series data underlying these models contain interaction terms (Ray et al., 2019, Schlenker and Roberts, 2009) or use QR decomposition (Ray et al., 2019) to lessen intra-term correlation bias and also commonly undergo de-trending (Heino et al., 2023, Vogel et al., 2019) procedures like singular spectrum analysis or (bi)linear interpolation (Heino et al., 2023, Schlenker and Roberts, 2009, Vogel et al., 2019) to increase the readability and scale alignment of data, respectively. Finally, cross validation (Heino et al., 2023, Ray et al., 2019, Schlenker and Roberts, 2009, Vogel et al., 2019), random training (Heino et al., 2023, Schlenker and Roberts, 2009) splits, and various statistical procedures (BIC (Ray et al., 2019), ANOVA F-Test (Ray et al., 2019), Gaussian Residuals (Ray et al., 2019), etc.) allow comparison of model efficacy after their creation.

It is generally agreed that anthropogenic climate change has and will continue to adversely affect crop yields. Models explain more—roughly 50%, of crop yield variability in higher, rainfed latitudes, and maize yields specifically were explained best these covariates (Heino et al., 2023). Also, extreme climate

events—especially for temperature, had a more significant effect (Schlenker and Roberts, 2009) on yields than mean—especially for precipitation, indicators. Divided by crop, yields have already changed between roughly 3% and 13% , but global losses are particularly destructive to $\frac{1}{2}$ of the countries with a ‘serious’ or greater hunger index, many of which are located in Africa or South Asia (Ray et al., 2019). Depending on the source, authors have found roughly between a 1% caloric loss (Ray et al., 2019) thus far or a 3.9% global yield loss (Heino et al., 2023) during years of particularly bad heat and drought from anthropogenic climate change. Overall, scholars agree that latitudes near the equator are to experience the greatest yield losses (Jägermeyr et al., 2021) over time—in particular for maize, while northern latitude crops like wheat have more ambivalent or even positive changes (Jägermeyr et al., 2021, Ray et al., 2019) before decreasing sharply in later decades as both average temperature and extreme temperature frequency increases.

1.2 Spatial impacts of nuclear winter

Nuclear winter’s impact on food security is spatially distinct—and generally more severe, from anthropogenic causes, and its impact also depends strongly on the size of the nuclear catastrophe considered. Most modern literature considers an exchange between India and Pakistan that results in 5 Tg of stratospheric soot injections (Jägermeyr et al., 2020, Mills et al., 2014, Robock et al., 2007), but Xia and Robock, among others, have considered up to 47 Tg between these parties and estimate up to 150 Tg to be involved in a nuclear exchange between the US and Russia (Xia et al., 2022). This is the rationale behind my choice to consider the 5 Tg and 47 Tg cases in my analysis. Irrespective of size, the stratospheric shortwave radiation absorption and heating (Mills et al., 2014, Robock et al., 2007) of emitted black carbon (BC), reflecting less light than particles from large past volcanic eruptions in Tambora (1815) or Laki (1783) (Robock et al., 2007, Xia et al., 2022), means that only a partial short-term rainout is possible and global effects are severe, especially in the more polar latitudes (Jägermeyr et al., 2020, Mills et al., 2014, Robock et al., 2007, Xia et al., 2022). In the 5 Tg case, for example, it’s estimated that a 1.2 – 1.6 Tg initial rainout is followed by lingering BC such that 1.1 Tg remains after 10 years (Mills et al., 2014), and authors estimate that the e-folding time of BC ranges from 6 (Robock et al., 2007) to 8.75 years. Fueled by advanced climate and crop models, climatic consequences include dramatic, slightly lagged (3 – 4 years) surface cooling, where global mean temperatures decrease by between 1.25 °C (Robock et al., 2007), 1.5°C (Mills

et al., 2014), and 1.8 °C (Jägermeyr et al., 2020) annually in the first 5 years and −0.5 °C a decade post-disaster in the 5 Tg case. For larger soot injections, this surface cooling can rise to 16 °C, and in either situation oceanic effects ensure damages could last up to 25 years (Xia et al., 2022, Mills et al., 2014). As for the effects described below, these impacts are inequitably distributed spatially, being most significant in poleward grain-growing latitudes with rainfed crops and large exporters. These areas project to see their growing season days without frost reduced by 10 – 20 days in the 5 Tg case alone (Robock et al., 2007). With respect to other climatic effects, shortwave radiation expects to decline by an average of 4.9% in the first 5 years post-disaster, corresponding roughly to a 10 – 15 W/m² maximum decrease here (Mills et al., 2014, Robock et al., 2007, Xia et al., 2022) in the 5 Tg case and up to 130 W/m² in the 150 Tg case. A smaller but similar annual 1.9% decline in longwave radiation is expected (Jägermeyr et al., 2020) over this time horizon, and the severity of these drops can be understood in that doubling atmospheric CO₂ increases shortwave radiation by 1.5 W/m². Similarly, annual precipitation declines very shortly after the conflict range from roughly −6% (Mills et al., 2014) to −8% (Jägermeyr et al., 2020) (5 Tg) to −68% globally (Jägermeyr et al., 2020, Xia et al., 2022), corresponding to a −0.3 mm d^{−1} to −2.0 mm d^{−1} global annual decline (Mills et al., 2014, Xia et al., 2022). Additional impacts include the expansion of sea ice, a 20 – 25% average global ozone loss—corresponding with stratospheric heating, a UV enhancement of 30 – 80% in mid-latitudes, and harmful potential UVI of 16 – 21 in parts of North America and Southern Europe (Mills et al., 2014).

Agricultural yields are similarly affected strongly with a focus on poleward latitudes. For example, Jägermeyr et al. (2020), among others, estimate a sharp global yield decline peaking at a single-year loss of 12%, quadruple the largest historical decline and primarily fueled by cooling temperatures. The same paper estimates a gradual recovery after 10 – 15 years, and average annual yield losses in the 5 years post-conflict are roughly 13% for maize, 11% for wheat, 3% for rice, and 17% for soybean. More recently, Xia and Robock predict a global 7% (5 Tg), 50% (47 Tg), 90% (150 Tg) calorie food production decrease annually in this same time horizon. This is again skewed towards the Northern Hemisphere, which could see reductions from 30% to 86% while lower latitudes may only see 10% depending on catastrophe weight. Because crop production is 24 times greater than fisheries and includes more calories per mass, these results are much damaging than decreases (3% for 5 Tg, 37% for 150 Tg) in fish net primary productivity (Xia et al., 2022). In this pa-

per, they also consider Livestock, Partial Livestock, and No Livestock scenarios post-conflict, and, depending on alterations in waste practices and assuming equitable food distributions, the most optimistic projections say caloric needs may even be met even under 47 Tg. However, under an exchange of 150 Tg, they add that most nations would have around $\frac{1}{4}$ of their citizens survive, but more realistically much smaller catastrophes would have comparably severe consequences due to domestic and local hoarding, income and power inequality, as well as other complicating factors (Xia et al., 2022). This is true especially considering they calculated earlier that the global stocks-to-use ratio (STU), an indicator of food reserves compared to domestic yearly use (Jägermeyr et al., 2020), was only 0.20, meaning most import-dependent nations in Africa or even some in northern latitudes will be worst off. That being said, these papers importantly assume exports cease post-conflict—an assumption I abandon in my analysis, so those with larger STUs like the US, Brazil, and Australia (Xia et al., 2022), and Argentina are expected to have smaller use declines because food is assumed to be reallocated to domestic use. However, global migration likely will worsen conditions in these states as well (Jägermeyr et al., 2020), painting a grim picture for all.

1.3 State vulnerabilities to food insecurity

Many authors continue to devise methods to approximate state vulnerability to food insecurity and starvation, many of whom directly or indirectly informed my construction of a post-disaster vulnerability score function. For instance, Fader et al. (2013) associate vulnerability with import dependency, which is taken as a function of the capacity of a state’s water and land resources, adjusted for conservation efforts, to fulfill food production requirements of their population level. Needed production is defined as imports plus domestic production minus exports (Fader et al., 2013), and the authors define metrics (WCI and CE) that, when exceeding 100% at various agricultural management (CUR, POT, HIG) and population levels, represent a nation that must be import-dependent (Fader et al., 2013), as all renewable, available, useful land/water is not sufficient. It was found that 66 countries (22 water boundary, 62 land boundary) are import-dependent, including 950 million people and spatially centered along mid-latitudes in northern African, Arabic, and Andean countries. Even at POT (i.e., optimal) land and water productivity and at all projected population levels (A2r, B1, B2), 30 of these nations are still import reliant, with similar spatial distributions as above (Fader et al., 2013). Notably, it is dually found that agricultural management is much more important to import depen-

dence quantity than population projections. Also, by 2050, depending on population change and economic development to improve agricultural productivity, 1.3 billion people in low-income economies and 5.2 billion globally may become import-dependent (Fader et al., 2013)—extremely concerning metrics considering that many of these nations import from northern latitude states particularly susceptible to nuclear winter yield declines.

Jehn et al. (2024) used alternative measures of vulnerability that are directly leveraged in my vulnerability score function. Their work is, to my knowledge, the first to rigorously abandon the post-disaster zero-export assumption and quantitatively assesses the trade impacts of nuclear catastrophe using weighted directed networks. Their analysis uses three important metrics that I apply for this purpose: Jaccard Distance, Community Satisfaction, and Node Stability (Jehn et al., 2024). Given both that these terms rely on a basic understanding of networks and because their literature is a core part of the technical foundation of my work, a more rigorous discussion of their methodology and metrics can be found in Section 2.2 and Section 2.5, respectively.

1.4 Networks and community detection

Because my work views international trade as a weighted, directed network and employs a particular community detection model, a basic foundation in both areas is necessary. Note that graph and network are used interchangeably in this paper.

Let $G(N, E, W)$ denote a graph with node set N , edge set E , and non-negative weight set W . Nodes are simply network entities (e.g., countries) while edges symbolize relationships (e.g., trade) between two entities. Edge weights, if distinct as to constitute a weighted graph, reflect relationship strengths between nodes and closely interconnected sections of nodes are called communities, from which a graph’s community structure is formed (Kojaku and Masuda, 2018). Let e_{ij} represent the edge between node i and node j of $G(N, E, W)$. Then, if $e_{ij} \neq e_{ji}$, then that edge is directed (i.e., has a direction). If the edges of $G(N, E, W)$ are of this form, then G is a directed graph. Let $G[S]$ be one such community formed from $S \subset N$ nodes.

Underlying community structure can reveal essential information about complex graph relationships, and as such many community detection methods are available. Some, like the Louvain and FastGreedy methods (He et al., 2019), optimize quality metrics such as modularity, which measures the number of edges in a community compared to what is expected of in an equivalent graph of randomly placed edges (Newman, 2006). This is the primary community detection

method employed by Jehn et al. (2024) in their post-disaster trade analysis. See Figure 1 for an example of a weighted, directed network for wheat trade with communities detected through modularity optimization.

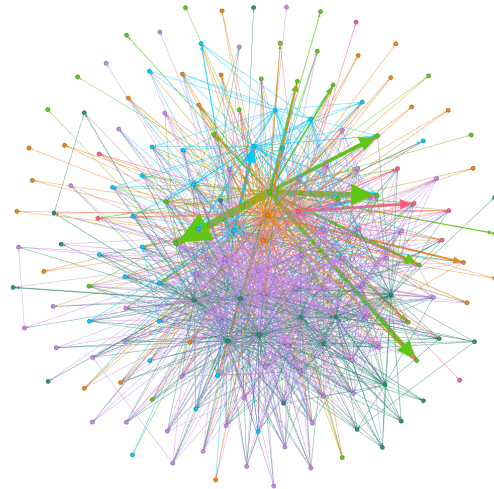


Figure 1: An example of a weighted, directed network of wheat trade, separated into colored communities by modularity optimization and Gephi’s Fruchterman Reingold layout. As in Jehn et al. (2024), nodes represent countries and directed edges represent net export wheat volume traded. Thicker edges equate to higher net volume.

Others have developed alternative sampling or analytical methods to determine the formal statistical significance of a graph. By this, I mean deriving a p-value for a single community like $G[S]$ under a random null model like the Erdős-Rényi (E-R) model, configuration model, Stochastic block model, or Poisson random model (He et al., 2019). The details of each are unimportant in this work, and there is no consensus on the best method for this task. Sampling methods often use the probability of finding a higher quality community of the same size and/or node set by random chance, although definitions of community quality vary (Kojaku and Masuda, 2018, He et al., 2019). Analytical methods may attribute higher significance to communities with many nodes of low membership probability in that community, again with respect to a random null model, although how this probability is quantified again varies. Some examples of this form include C-Score, SSR, ESSC, OSLOM, and CCME, of which only the last two pertain to weighted graphs (He et al., 2019). Alternatively, they may investigate the community itself more directly, as performed by He et al. (2021) in

their Logrank test. It is the direct nature of this method, paired with its allowed use in weighted, directed graphs, that makes it the method of choice for my trade community statistical significance procedure.

2 Methods and data

Developing my analysis of international post-catastrophe food security through community significance and state vulnerability relies heavily on both the technical network framework of Jehn et al. (2024) as well as the community statistical significance methodology generated by He et al. (2021). I provide a detailed explanation of how I combine these pieces of literature.

2.1 Abandoning the zero-export assumption

After a nuclear catastrophe of any weight, food security is commonly measured under a zero-export model, where each state relies entirely on its domestic food production for survival (Jägermeyr et al., 2020, Xia et al., 2022). The rationale behind this assumption is clear, as crisis hoarding practices have been observed in previous catastrophes nowhere close to as deadly. Also, even if international trade continued in some form, it is difficult to accurately predict in what forms it may manifest, especially if the combatants in the nuclear catastrophe are unknown. However, it is equally unrealistic to assume that, given the large number of already import-dependent states detailed in Section 1.3, post-catastrophe trade network modeling has no value, as even simple analysis can bring clarity to policymakers in vulnerable or high-exporting states that may help reduce post-disaster inefficiencies associated with hoarding food and ultimately promote the well-being of a greater number of individuals. At worst, even imprecise frameworks can direct further future research

2.3 The logrank test for community detection

With trade communities generated, I employ the Logrank test detailed by He et al. (2021) to evaluate each of their statistical significance. To reiterate, this method is chosen because of its unique ability directly evaluate whole communities compared to its peers (i.e., OSLOM, CCME) with the capacity to work with weighted networks. Additionally, the Logrank test was found to be significantly better than conductance, modularity, and CCME methods under the Nemenyi and Bonferroni-Dunn tests for weighted networks (He

et al., 2021), making it both a simple and effective choice.

2.2 International trade as a weighted, directed network

Before I could evaluate the post-disaster statistical significance of trade communities, I relied extensively on the open-source framework provided by Jehn et al. (2024) here (<https://github.com/allfed/pytradeshifts>).

Using third-year post-disaster yield loss data from Xia et al. (2022) and annual production and import FAO data for trading partners, I created networks to represent international trade for each considered crop and for each catastrophe weight. As in Jehn et al. (2024), nodes represented states and edges were the directed, weighted net export volume between two states—corrected for re-exports to only consider point-of-origin to point-of-destination flows. Like Jehn et al. (2024), I exclude trade flows below the 75th percentile to focus on main flows. However, since my focus is nuclear winter, I only consider the pre-disaster case and their Abrupt Sunlight Reduction Scenario (ASRS)—equivalent to nuclear winter, in my analysis. To apply this scenario, trade flows are scaled proportionately by each state’s percentage yield loss to generate new net export edge weights.

From here, I use the Louvain method from NetworkX to generate trade communities for each crop and for each scenario. In this way, trade communities are thought of as states that trade amongst themselves with greater comparative volume, making them a good measure of primary partners. These are the communities that will be evaluated for statistical significance. An example community visualization from Jehn et al. (2024) for ASRS wheat is provided in Figure 2 and 3.

et al., 2021), making it both a simple and effective choice.

Consider arbitrary trade network $G(N, E, W)$ for some scenario and crop. This method considers both present and absent (i.e., zero weight) edges as censored observations, so I first generate E' such that it considers unweighted edges too. Let A be the set of unweighted edges. Therefore, $E' = E \cup A$. I modify each network using NetworkX such that these zero weight edges are present, forming graph $G(N, E', W)$ in this case.

Following He et al. (2021) directly in Python, let $G[S]$ be one trade community with nodes $S \subset N$ generated by the Louvain method on $G(N, E, W)$. For this

Trade communities for wheat with base year 2022

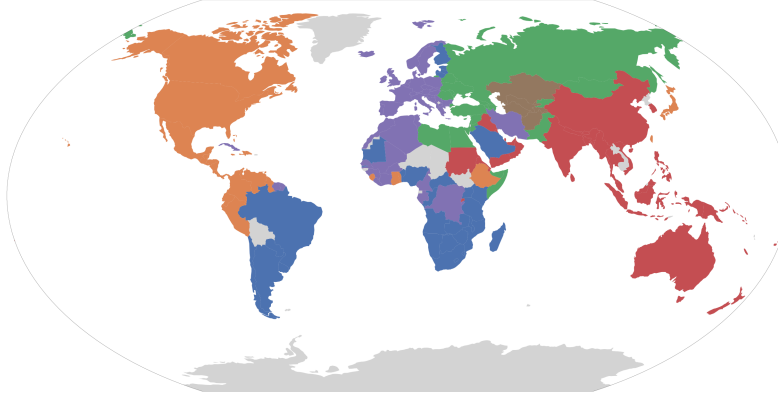


Figure 2: Trade communities generated under pre-disaster conditions. Each color represents a trade community of states. This graphics were originally generated by [Jehn et al. \(2024\)](#).

Trade communities for wheat with base year 2022
in scenario: Abrupt Sunlight Reduction Scenario

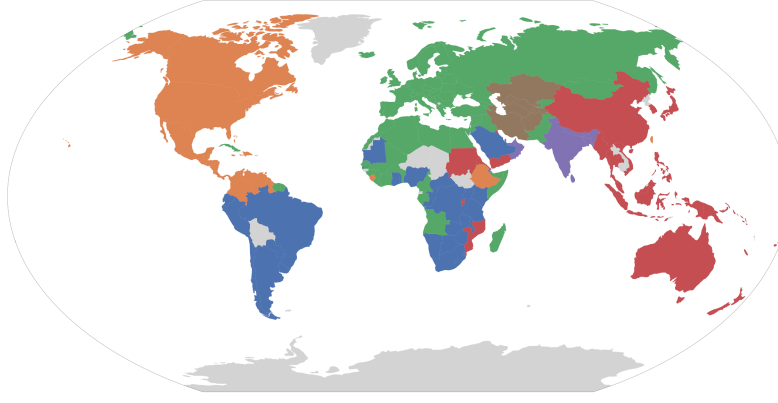


Figure 3: Trade communities generated after a 47 Tg nuclear catastrophe. Each color represents a trade community of states. This graphics were originally generated by [Jehn et al. \(2024\)](#).

community, the internal edge weight set W_S^{in} refers to edges of E' between two nodes of S , while the external edge weight set W_S^{out} refers to edges of E' between one node of S and another node of $N \setminus S$. Let $F_{in}(\cdot)$ be the complementary cumulative distribution function considering the internal edge weight set and their respective internal frequencies, and let $F_{out}(\cdot)$ be the complementary cumulative distribution function considering the external edge weight set and their respective external frequencies.

Then, as explained in [He et al. \(2021\)](#), the louvain method is a non-parametric two sample test as follows:

$$H_0 : F_{in}(w) = F_{out}(w), \quad \text{for all } w \geq 0. \quad (1)$$

$$H_1 : F_{in}(w) > F_{out}(w), \quad \text{for at least one } w \geq 0. \quad (2)$$

Here, $w \in W_S$, where $W_S = W_S^{in} \cup W_S^{out}$. This pro-

cess makes intuitive sense, as significant communities would possess a greater proportion of internal higher-weight edges reflective of a complementary CDF that significantly greater than that of external edge weights for some w . For each weight w , 3x3 contingency tables of terms as produced exactly following [He et al. \(2021\)](#). These tables are used to produce the Logrank test statistic Z , which, under H_0 , is approximately standard normal. Knowing this, the community p-value is easily generated. Please see both [He et al. \(2021\)](#) if interested in the precise test procedure for directed, weighted networks and [Tian](#) for more detail on the Logrank test.

For each Louvain method community generated for every crop, scenario (i.e., base and ASRS), and catastrophe weight (i.e., 5 Tg and 47 Tg), these p-values were stored. I use an significance level of 0.01.

2.4 Multiple testing adjustment

Because the above hypothesis test had to be performed for each community for every crop, scenario, and catastrophe weight, my unadjusted p-values have a multiple testing problem. To solve this, I employ the Bonferroni-Holm correction procedure (i.e., Holm step-down approach) twice, once for each set of unadjusted p-values at a particular catastrophe weight. A precise description of this method is given in [Simon \(2005\)](#). I choose this method to avoid over-conservative nature of the Bonferroni correction while still maintaining relative simplicity. These adjusted p-values are different from what is performed by [He et al. \(2021\)](#) and are used for my community significance results.

2.5 Building a vulnerability function

With all Louvain method trade community p-values generated, I now shift towards building a function that approximates individual state post-catastrophe vulnerability to food insecurity. For this purpose, I devise the following function $V_d : \mathbb{R}^8 \rightarrow \mathbb{R}$ such that:

$$V_d(v) = \sum_{i \in T} [A j_{d,i} c_{d,i} - (B y_{d,i} p_{d,i} + C I_{d,i} r_{d,i} + D \Delta n_{d,i} + E \Delta s_{d,i})]$$

Here, set $T = \{\text{wheat, rice, soybeans, maize}\}$, d refers to a particular country, and v is a vector of the following variables, excluding the last coefficients:

- $j_{d,i}$: The Jaccard Distance between the base and post-catastrophe trade communities of country d for crop i , scaled by 100. Here, $j_{d,i} \in [0, 100]$.
Let C_1 and C_2 be the base and post-catastrophe trade communities. Then, from [Jehn et al. \(2024\)](#), Jaccard Distance, denoted here $d_J(C_1, C_2)$, measures the extent of dissimilarity between these sets. Precisely, $d_J(C_1, C_2) = 1 - \frac{|C_1 \cap C_2|}{|C_1 \cup C_2|}$, where $|\cdot|$ is the standard counter. Thus, $d_J(C_1, C_2) \in [0, 1]$, where values closer to 1 indicate greater distance.
- $c_{d,i}$: A boolean equaling 1 if the post-catastrophe community of country d is statistically significant, 0 otherwise. Here, $c_{d,i} \in \{0, 1\}$.
- $y_{d,i}$: Percentage country-level yield relative change between the base and nuclear catastrophe scenarios for crop i , scaled by 100. See Section 2.3 in [Jehn et al. \(2024\)](#) for computation details. Here, $y_{d,i} \in [-100, 100]$, where -100 indicates the largest yield decline.
- $p_{d,i}$: The pre-disaster production rank for crop i . Here, $p_{d,i} \in \{1, 2, 3, 4\}$, where 4 is assigned to the most produced crop (1 to the least) in country d .

- $I_{d,i}$: Percentage country-level import relative difference between the base and nuclear catastrophe (ASRS) scenarios for crop i , scaled by 100. See [Jehn et al. \(2024\)](#) or their aforementioned github for computation details. Here, $I_{d,i} \in [-100, 100]$, where -100 indicates the largest import decline.
- $r_{d,i}$: The pre-disaster imports rank for crop i . Here, $r_{d,i} \in \{1, 2, 3, 4\}$, where 4 is assigned to the most imported crop (1 to the least) in country d .
- $\Delta n_{d,i}$: Percentage change in standardized Node Stability for country d and crop i between the pre-disaster base and nuclear catastrophe scenarios, scaled by 100. Node Stability refers to the relative ease by which country d can replace their import partners, capturing factors like spatial distance, volume traded, and political stability. See Section 1.2.2 in [Jehn et al. \(2024\)](#) and Section 2.1.2 in [Wang et al. \(2023\)](#) for more details. Here, $\Delta n_{d,i} \in [-100, 100]$.
- $\Delta s_{d,i}$: Percentage change in standardized Community Satisfaction for country d and crop i between the base and nuclear catastrophe (ASRS) scenarios, scaled by 100. Community Satisfaction is the sum of trade community imports divided by the sum of all imports to country d , capturing the capacity of a trade community to fulfill its import needs. See Section 1.2.1 in [Jehn et al. \(2024\)](#) for more details. Here, $\Delta s_{d,i} \in [-100, 100]$.
- A, B, C, D , and E are arbitrary scalars all equal to 1 in my analysis. Scalars can be changed in future analysis to reflect different opinions about which terms are most crucial to a state's vulnerability to post-disaster food insecurity.

For clarity, $V_d(v)$ was designed such that larger, positive scores correspond to greater state vulnerability. Term by term, $j_{d,i} c_{d,i}$ attributes higher scores to states whose trade communities are both statistically significant and strongly different post-catastrophe, as states who we are confident will be forced to rely most heavily on different partners after a disaster are subject to the most severe transition vulnerabilities.

Next, $-y_{d,i} p_{d,i}$ and $-I_{d,i} r_{d,i}$ account for ranked losses in both domestic yield and international imports, two essential components of a state's food security. The ranking system is used over alternative weighting schemes (e.g., production volume adjusted for population) as to weight heavily more important crops whilst not creating additional complexities.

Then, $-\Delta n_{d,i}$ captures the risk of total state destruction after a nuclear catastrophe. For example, if a

country d cannot afford to replace a particular node in their post-disaster trade community—and that node is destroyed by nuclear weapons, then country d is much worse off. This term accounts for this case and more generally penalizes fragile trade communities.

Finally, $-\Delta s_{d,i}$ captures the logistical challenges that will present themselves after a nuclear catastrophe. In particular, this term penalizes states who become increasingly unable to receive import needs from their immediate trade community, as such states are unlikely to have the post-disaster infrastructure to maintain a complex trade network and are subsequently increasingly vulnerable to food insecurity and starvation.

With this construction, I compute and rank vulnerability scores for every country into two separate tables, one for the 5 Tg case and the other for the 47 Tg case. I then standardize scores such that their units, which are relatively arbitrary, have greater meaning. This permits stronger comparisons between states within and between each table.

3 Results

With respect to community significance between 5 Tg and 47 Tg catastrophe weights, I find gener-

ally that the percentage of significant trade communities across crops increases from 71.739%, or $\frac{33}{46}$, to 81.633%, or $\frac{40}{49}$, signaling that most trade communities identified by the Louvain method were significant at either disaster weight. The adjusted p-values for all communities at each weight can be found in the Appendix. Although I have not tested if this percentage increase is significant, it may be explained by the fact that larger catastrophe weights more completely destroy international yield and import volume—lessening trade complexity to a few exporting exceptions and ensuring the complementary CDF of internal edge weights for identified communities around those exceptions are more easily recognizable. This loss of complexity is corroborated by the fact that, at either weight, I found more insignificance amongst base communities.

Broadly, I also find that soy and especially rice communities tend to be the most insignificant across weights, which may be explained simply by greater uniformity of net export volume amongst rice and soy trading partners. This is distinct from wheat and maize, for example, where their trade relies on only a few extremely large exporters. An example of the spatial distribution of rice community significance in the 47 Tg is given in Figure 4 and 5.

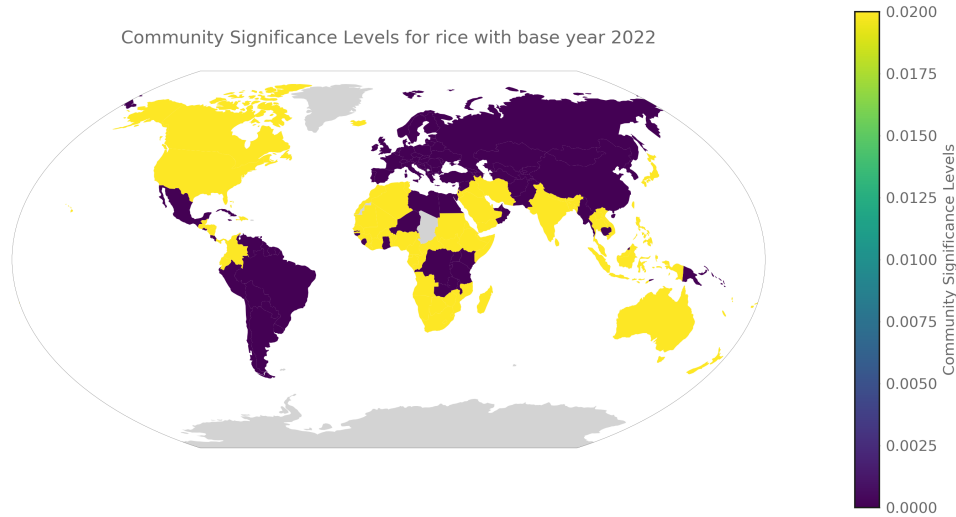


Figure 4: Distribution of rice community significance in the base scenario. Lighter colors pertain to less significant communities.

Knowing that post-disaster yield impacts project the least damage to rice, especially under smaller weights, its greater post-disaster community insignificance and corresponding food trade uncertainties are slightly less concerning. For policymakers, however,

lacking confidence in post-disaster trade means less preventative work can be done.

Generally, my vulnerability score function finds that the spatial distribution of most vulnerable states shifts from northern Europe to central Asia with in-

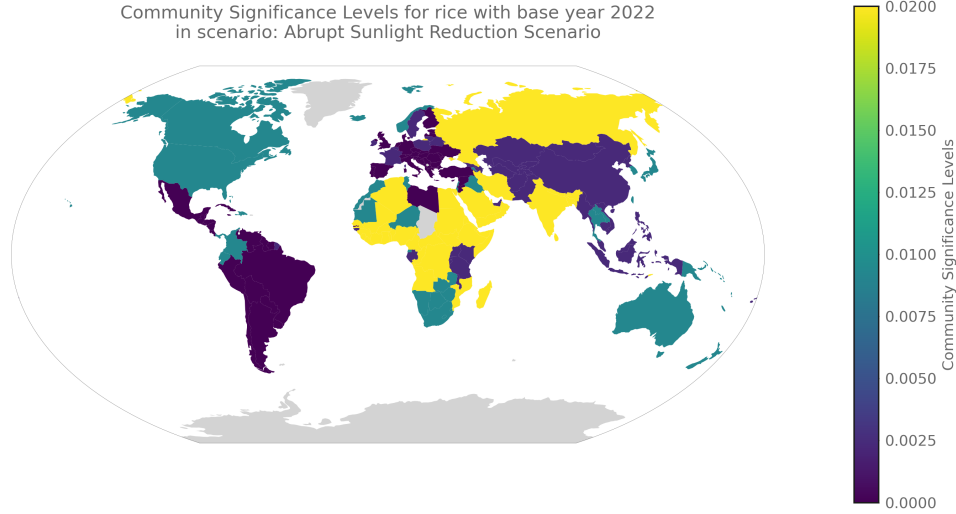


Figure 5: Distribution of rice community significance in the 47 Tg ASRS scenario. Irrespective of case, no spatial community significance relationships are found. Lighter colors pertain to less significant communities.

creasing catastrophe weight whilst the scores themselves more than double. Despite this, high ranking, shared vulnerable states between 5 Tg and 47 Tg weights, respectively, include Sweden (rank 1 and 4), Kazakhstan (rank 12 and 5), Ukraine (rank 32 and 1), Finland (rank 2 and 9), Canada (rank 5 and 8), Mongolia (rank 4 and 16), Switzerland (rank 7 and 2), Tajikistan (rank 18 and 3), and Azerbaijan (rank 14 and 6). Notable others include the United States (rank 11 and 20) and Russia (rank 13 and 19).

At the same time, the standardized scores of states most vulnerable in the 5 Tg scenario are much larger and themselves have higher variance—a testament to the more spatially uniform yield and import destruction of larger nuclear disasters. Below I present more specific details about both these significance and vulnerability findings.

3.1 Community significance: 5 Tg case

At this catastrophe weight, I find that most insignificant communities were limited to soybean (2 base, 5 ASRS) and rice (3 base, 3 ASRS), while wheat (1 base, 1 ASRS) and maize (0 base, 0 ASRS) had comparatively less. The members of such insignificant communities were likely to be shared between base and ASRS (post-catastrophe) scenarios, which is explained by the fact that most country Jaccard Distances, especially for wheat and maize, were close to 0. For isolated communities with higher member Jaccard Distances—often soy or rice, they were mostly also insignificant, casting doubt on the severity, impact or disruptions of such post-disaster trade alterations. See the Appendix for a more comprehensive overview of in-

significant communities.

Looking at Figure 6, given that the strongest yield declines at this weight project to be maize, wheat, and soybean, stronger trade community significance for maize and wheat presents opportunities for policymakers to have greater clarity for the most important post-disaster food relationships and strengthen ties between included countries.

3.2 Community significance: 47 Tg case

At this catastrophe weight, I find that the most insignificant communities were again limited to soybean (2 base, 1 ASRS) and rice (3 base, 1 ASRS), while wheat (1 base, 0 ASRS) and maize (0 base, 1 ASRS) again had comparatively less. Unlike before, the members of such insignificant communities were unlikely to be shared across scenarios, as evidenced by roughly the middle 50% of post-disaster trade communities having Jaccard Distances between 0.4 and around 0.8, with only few communities across crops with Jaccard Distances at or below 0.2. Unlike before, states with the greatest Jaccard Distances often paired with the most significant communities, suggesting both vulnerability for such states and potential for policymaker impact stemming from these well-defined post-disaster communities. Again, see the Appendix for a more comprehensive overview of insignificant communities.

Looking at Figure 7, given that the heaviest yield impacts at this weight still project to be maize, wheat, and soybean, stronger trade community significance for maize, wheat, and even soybean again presents opportunities for policymakers to have greater post-disaster clarity and strengthen relevant relationship ties.

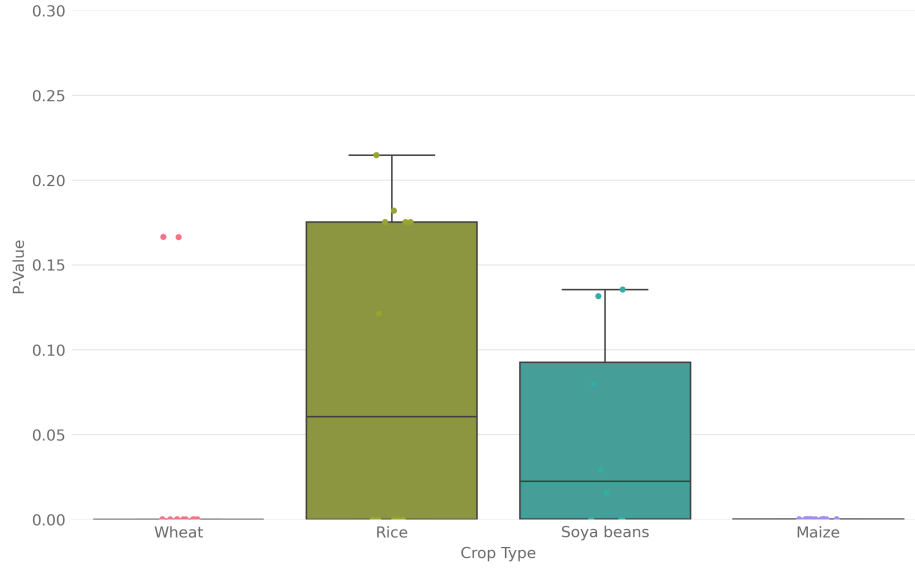


Figure 6: A boxplot including the base and 5 Tg ASRS scenario adjusted community p-values by crop type. The median adjusted p-value for both rice and soy exceeds the 0.01 significance level. For rice, the median even exceeds a 0.05 significance level, suggesting greater uncertainty for post-disaster trade relationships related to these crops.

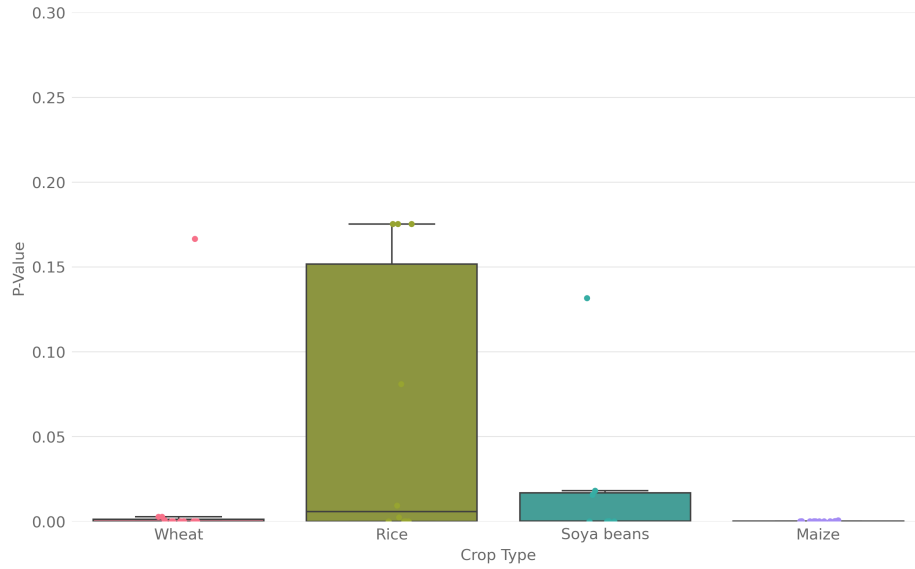


Figure 7: A boxplot including the base and 47 Tg ASRS scenario adjusted community p-values by crop type. The median adjusted p-value for all crops is less than my 0.01 significance level. However, the third quartile for rice, which includes three communities with p-value of 0.1752, pushes its boxplot well above the both the 0.01 and 0.05 significance level. Here, this rice community insignificance presents more concern than for the 5 Tg scenario with respect to trade network stability given that both yield and import declines are strong and comparable to other crops at this weight. The p-value range for both of these charts were manually limited at 0.30, despite some outliers.

3.3 State vulnerabilities: 5 Tg case

Here, I deconstruct vulnerability scores for top states in the 5 Tg case to determine their driving fac-

tors. I find that the $-y_{d,i}p_{d,i}$ term drives leading scores for the most vulnerable states for food insecurity.

- **Sweden:** The $-y_{d,i}p_{d,i}$ term contributes 79.23%,

or $\frac{593.46}{749.08}$, to its vulnerability score. It is especially harmed by its wheat (163.24) and maize (230.64) contributions.

- **Finland:** The $-y_{d,i}p_{d,i}$ term contributes 77.90%, or $\frac{477.22}{612.60}$, to its vulnerability score, led by wheat (190.88). Notable here is that all other function factors for wheat are negative, and I also find a large contribution from the $-I_{d,i}r_{d,i}$ term for maize (148.88).
- **Canada:** The $-y_{d,i}p_{d,i}$ term contributes 89.36%, or $\frac{418.55}{468.37}$, to its vulnerability score, led by maize (199.83). Notable here is the large $-I_{d,i}r_{d,i}$ term for maize (60.87) as well as a strong increase in soybean Community Satisfaction (-70.82).
- **Ukraine:** The $-y_{d,i}p_{d,i}$ term contributes 41.14%, or $\frac{101.25}{224.28}$, to its vulnerability score, led by soybeans (62.36) and maize (50.08) and interestingly benefited by wheat (-41.37) contributions. This percentage is comparatively smaller than its peers, which may have contributed to its much lower ranking (32) than in the 47 Tg scenario (1). I also find that the $-I_{d,i}r_{d,i}$ term contributes 41.72%, or $\frac{93.57}{224.28}$ to its score, notably hampered by a strong improvement in post-disaster soybean ranked imports (-75.90).
- **Switzerland:** The $-y_{d,i}p_{d,i}$ term contributes 66.86%, or $\frac{237.29}{355.05}$, to its vulnerability score, led by maize (172.14) and soybean (28.70). The $-I_{d,i}r_{d,i}$ term contributes 25.62%, or $\frac{90.96}{355.05}$, to

its score, notably hampered by a strong improvement in soybean ranked imports (-75.90).

- **Kazakhstan:** The $-y_{d,i}p_{d,i}$ term contributes 63.24%, or $\frac{209.88}{331.89}$, to its vulnerability score, led by rice (78.32) and maize (73.32). The $-I_{d,i}r_{d,i}$ term contributes 33.80%, or $\frac{112.19}{331.89}$, to its score.
- **Mongolia:** The $-y_{d,i}p_{d,i}$ term contributes 66.37%, or $\frac{315.94}{476.04}$, to its vulnerability score, led by wheat (105.00) as well as both rice and soy (79.10). Like Kazakhstan, it sees improvement in wheat ranked imports: -11.34 for Mongolia and -15.13 for Kazakhstan. However, this is small and both have insignificant post-disaster wheat trade communities, casting doubt on confidence in these results. The $-I_{d,i}r_{d,i}$ term contributes 23.57%, or $\frac{133.32}{476.04}$, to its score, led by soybean (66.39).
- **United States:** The $-y_{d,i}p_{d,i}$ term contributes 32.93%, or $\frac{111.01}{337.20}$, to its vulnerability score, led by maize (60.88) primarily. Interestingly, the $-I_{d,i}r_{d,i}$ term is more important, contributing 55.69%, or $\frac{187.80}{337.20}$, to its score and being led by wheat (81.62).
- **Russia:** The $-y_{d,i}p_{d,i}$ term contributes 83.33%, or $\frac{276.03}{331.23}$, to its vulnerability score, led by maize (154.92) primarily. Notably, we see that ranked wheat yields increase marginally (-15.12). The $-I_{d,i}r_{d,i}$ term contributes 21.00%, or $\frac{69.56}{331.23}$, to its score and is led by rice (33.81).

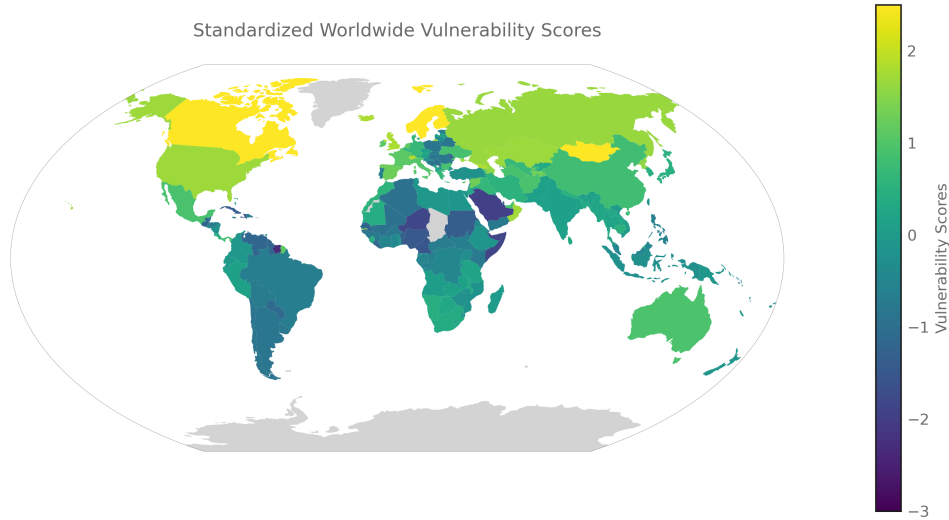


Figure 8: A map of standardized state vulnerability scores in the 5 Tg scenario. Lighter colors correspond to greater state vulnerability to post-disaster food insecurity. Again, it is clear that northern latitude states tend to constitute the most vulnerable. Asymmetric, spatially dependent disaster impacts mean ranked yield declines contribute most.

For this weight, I find that the top five leaders in standardized vulnerability—Sweden, Finland, Norway, Mongolia, and Canada, all have standardized scores in excess of the standardized leader in the 47 Tg case, suggesting that smaller disaster weights encourage greater spatial impact variability. Also, we see that some state traditional nuclear risks like India and Pakistan are not nearly as vulnerable as those in the northern European latitudes, corroborating with the aforementioned literature. See Figure 8 for a spatial distribution of state food insecurity vulnerability and the Appendix for a more comprehensive list of vulnerable states at this weight.

3.4 State vulnerabilities: 47 Tg case

Here, I deconstruct vulnerability scores for top states in the 47 Tg case to determine their driving factors. I find that the $-I_{d,i}r_{d,i}$ holds relatively greater importance here than in the 5 Tg scenario.

- **Sweden:** The $-y_{d,i}p_{d,i}$ term contributes 43.75%, or $\frac{900.00}{2056.97}$, to its vulnerability score. This is a much smaller percentage than in the 5 Tg scenario, and the major contributors are wheat (400.00) and maize (300.00). The $-I_{d,i}r_{d,i}$ term contributes similarly at 41.22%, or $\frac{847.95}{2056.97}$, to its score, led by maize (298.88) and rice (275.16). Interestingly, three-fourths of the $j_{d,i}c_{d,i}$ terms are relatively large and significant, and it sees large declines in Node Stability.
- **Finland:** The $-y_{d,i}p_{d,i}$ term contributes 35.72%, or $\frac{700.00}{1959.73}$, to its vulnerability score, led by wheat (400.00). This is again a noticeably smaller percentage contribution than in the 5 Tg case, suggesting that the strong uniform yield declines, being less spatially dependent in the 47 Tg case, contribute to ranked import declines holding more influence over vulnerable states. The $-I_{d,i}r_{d,i}$ term contributes more at 44.94%, or $\frac{880.00}{1959.73}$, to its score. Finally, we see here a markedly higher $j_{d,i}c_{d,i}$ term here for wheat (93.68 vs. 0.00) and maize (74.47 vs. 4.26) than in the 5 Tg scenario.
- **Canada:** The $-y_{d,i}p_{d,i}$ term contributes 50.84%, or $\frac{998.24}{1963.44}$, to its vulnerability score, now led by wheat (398.24) and maize (300.00). Notably, the $-I_{d,i}r_{d,i}$ term contributes 41.37%, or $\frac{812.24}{1963.44}$, and it has strong improvement in the $-\Delta s_{d,i}$ term for soybean (−84.12). Finally, it has some of the most extreme losses of the $-\Delta n_{d,i}$ term for maize (52.56) and soybean (60.00) overall.
- **Ukraine:** The $-y_{d,i}p_{d,i}$ term contributes 39.09%, or $\frac{856.84}{2192.17}$, to its vulnerability score, led by maize

(400.00), soybean (200.00), and wheat (156.84). The $-I_{d,i}r_{d,i}$ term contributes 34.37%, or $\frac{753.5}{2192.17}$ to its score, driven by rice (253.50) and maize (300.00). Interestingly, the large ranking increase here is likely explained by the extremely large $-\Delta s_{d,i}$ scores for maize (100) and soybean (100), indicating total loss of Community Satisfaction compared to their 5 Tg values of 0.00.

- **Switzerland:** The $-y_{d,i}p_{d,i}$ term contributes 45.08%, or $\frac{936.73}{2078.10}$, to its vulnerability score, led by wheat (342.76) and maize (300.00). The $-I_{d,i}r_{d,i}$ term contributes 36.47%, or $\frac{757.95}{2078.10}$, to its score, driven again mostly by maize (296.49) and wheat (248.03). Notably, it has a large $j_{d,i}c_{d,i}$ term for all crops (e.g., soy is 84.84), especially compared to those in the 5 Tg scenario near 0.00, suggesting greater confidence in trade transition severity.
- **Kazakhstan:** The $-y_{d,i}p_{d,i}$ term contributes 45.45%, or $\frac{928.00}{2041.99}$, to its vulnerability score, led primarily by wheat (329.00), maize (299.61), and rice (200.00). The $-I_{d,i}r_{d,i}$ term grows in importance here to a 42.51%, or $\frac{868.04}{2041.99}$, score contribution. This is attributed to large wheat (379.95) and rice (212.92) contributions for this factors explained by many less imports from its central Asian community partners. Also, the $j_{d,i}c_{d,i}$ term for soybean (94.44) and rice (68.97) are very large compared to the 5 Tg case, suggesting greater transitional worry for Kazakhstan at this weight. Finally, it sees a strong gain in Community Satisfaction and strong loss in Node Stability for maize equivalently represented as a change of −68.11 for $-\Delta s_{d,i}$ and 43.00 for $-\Delta n_{d,i}$.
- **Mongolia:** The $-y_{d,i}p_{d,i}$ term contributes 38.10%, or $\frac{700.00}{1837.38}$, to its vulnerability score, led mostly by wheat (400.00). Again, the $-I_{d,i}r_{d,i}$ term grows in importance to contribute 50.59%, or $\frac{929.53}{1837.38}$, to its score. Notable also is its large $j_{d,i}c_{d,i}$ terms for wheat (73.97) and rice (68.97) as well as its maize changes in $-\Delta s_{d,i}$ and $-\Delta n_{d,i}$, the former for being an improvement (−26.14) and the latter being a decline (35.43) like for Kazakhstan.
- **United States:** The $-y_{d,i}p_{d,i}$ term contributes 45.07%, or $\frac{806.941}{1790.30}$, to its vulnerability score, again led by maize (373.96) and soybean (230.70) primarily. The $-I_{d,i}r_{d,i}$ term is similarly important, contributing 43.80%, or $\frac{784.22}{1790.30}$, to its score and being again led by wheat (390.48). Notable also is its strong loss in wheat

and maize Community Satisfaction, representing $-\Delta s_{d,i}$ values of 79.08 and 42.43, respectively. Finally, it experiences mixed change in Node Stability for wheat and maize, representing $-\Delta n_{d,i}$ values of 38.50 and -30.83 , respectively.

- **Russia:** The $-y_{d,i}p_{d,i}$ term contributes 54.57%, or $\frac{979.96}{1795.85}$, to its vulnerability score, led by near complete losses for wheat (379.96), maize (300.00), soybean (200.00), and rice (100.00). The $-I_{d,i}r_{d,i}$ term contributes less with 34.26%, or $\frac{615.20}{1795.85}$, towards its score and is led by soybean (207.89) and again rice (185.97).

For this weight, I find that the top five leaders in standardized vulnerability—Ukraine, Switzer-

land, Tajikistan, Sweden, and Kazakhstan, have much more similar scores, suggesting greater uniformity—but worse severity, in vulnerability with higher weight nuclear disasters. Here, these vulnerable states are increasingly focused on central Asia rather than northern Europe, but generally still favor northern latitudes. Interestingly, as this slightly contradicts existing literature, Australia is still relatively vulnerable, earning standardized scores of 0.942 and 0.888 between the 5 Tg and 47 Tg scenarios, respectively. See Figure 9 for a spatial distribution of state food insecurity vulnerability and the Appendix for a more comprehensive list of vulnerable states at this weight.

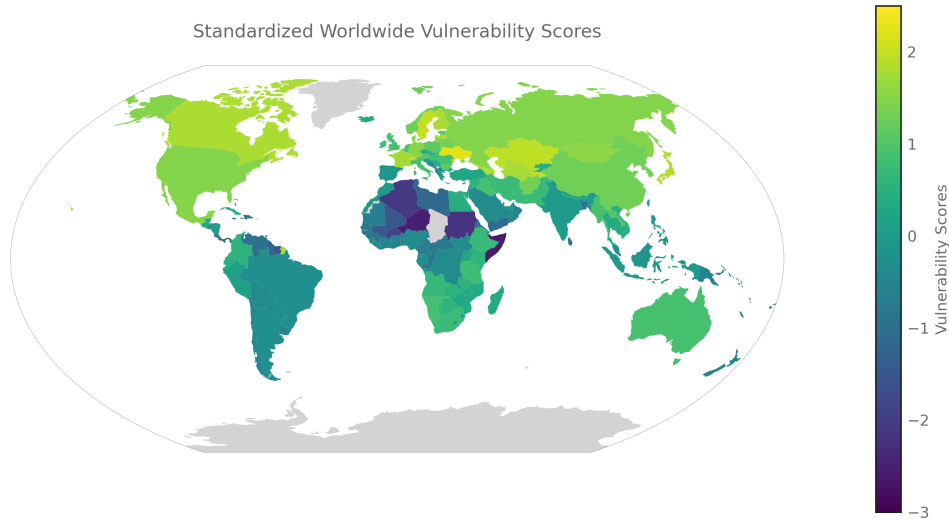


Figure 9: A map of standardized state vulnerability scores in the 47 Tg scenario. Lighter colors correspond to greater state vulnerability to post-disaster food insecurity. Again, it is clear that northern latitude states tend to constitute the most vulnerable, although individual standardized scores are less extreme and variable.

4 Discussion

I have presented an comprehensive approach that builds off of the works of [Jehn et al. \(2024\)](#) and [He et al. \(2021\)](#) to develop a more statistically rigorous and nuanced outlook on post-catastrophe trade relationships while identifying key state vulnerabilities to food insecurity and starvation. As one of the first reports to investigate the consequences of abandoning the zero-export assumption, many perhaps unrealistic simplifications were made in my analysis to serve as a starting point for future research. Nevertheless, I am confident that state policymakers can find value from this report to introspectively evaluate the fragility of their trade networks and vulnerability to post-disaster food insecurity.

4.1 Research gaps and future improvement

Many measures were taken in my analysis that provides room for future improvement. For example, I only consider the statistical significance of a community with respect to the edge distributions present in its own network. However, I fail to use statistical means to evaluate how different two communities are before and after a nuclear disaster, relying instead on Jaccard Distance. Moreover, my analysis only considers the 5 Tg and 47 Tg catastrophe scenarios, failing to investigate alternative weights or create an overall vulnerability score that captures how likely each weight catastrophe is to occur and their respective states consequences.

Also, my analysis ignores political, economic, or social factors that may complicate the simple relationship I use between national yield decline and net exports to adjust trade networks after a nuclear catastrophe. Finally, future improvement could take the form of using different statistically rigorous community detection and p-value adjustment methods outside of the Logrank test and Bonferroni-Holm correction, for example. Irrespective of research gap, I am confident future research can and will build on my analysis.

4.2 Conclusion

In this report, I visualize international food trade as a weighted, directed network and use a robust community detection test to evaluate the statistical significance of trade communities chosen by the Louvain method before and after 5 Tg and 47 Tg nuclear catas-

trophes. This was done to provide policymakers greater clarity both to the fragility of their food networks (for insignificant post-disaster communities) after nuclear catastrophe and to unforeseen trade partner transitions (for significant post-disaster communities and large Jaccard Distances) that, if prepared for, can save countless lives. Then, I build a thorough vulnerability score function that ranks states at greatest risk to food insecurity to simply alert state leaders that their country could be particularly unprepared for disaster. In doing so, I hope that preventative, deescalating, and/or logistical measures can be taken to again prevent worst-case estimates for loss of life in a nuclear catastrophe.

For a catastrophe in which death from bombs is just the beginning, it is as important as ever to make sure we stay aware of its cascading effects and prevent a nuclear disaster of any size from ever occurring.

5 Appendix

Wheat	Rice	Soya beans	Maize
0	0	1.8833617e-7	0
0	0	1.9219091e-7	0
0	1e-14	0.00012626466992	0
0	3.95e-12	0.01553308099871	0
1.2651561e-7	3.53107e-9	0.0291744953466	0
1.6543012e-7	3.53107e-9	0.07971075811282	0
4.6050811e-7	0.12122916378845	0.13154557875047	3.01e-12
4.9362342e-7	0.17524008784233	0.13546502875768	3.06e-12
0.00000578326767	0.17524008784233	null	0.00001655438969
0.00006394433719	0.17524008784233	null	0.0000338628403
0.16636368144586	0.18193923487342	null	0.0000850270796
0.1664708255361	0.21468704085556	null	0.00013617591681
null	null	null	0.00019569521423
null	null	null	0.00019569521423

Figure 10: An overview of adjusted community p-values for base and 5 Tg ASRS scenarios. Note that maize has the most trade communities, all significant at the 0.01 level, while soy had the least.

Wheat	Rice	Soya beans	Maize
0	0	2.82e-12	0
0	0	1.11585e-9	0
0	3.95e-12	1.8833617e-7	0
2.11495e-9	3.53107e-9	0.00012626466992	0
1.2651561e-7	0.00231982871994	0.01553308099871	3.01e-12
4.6050811e-7	0.00925986970698	0.01801533785238	8.38e-12
0.00000710902497	0.0808678694892	0.13154557875047	7.28533e-9
0.00006394433719	0.17524008784233	null	9.903272e-8
0.00070095579819	0.17524008784233	null	9.334333e-7
0.00264260049084	0.17524008784233	null	0.00001655438969
0.00264260049084	null	null	0.0000850270796
0.1664708255361	null	null	0.00013617591681
null	null	null	0.00056617852728
null	null	null	1

Figure 11: An overview of adjusted community p-values for base and 47 Tg ASRS scenarios. Note that maize has the most trade communities, all but one significant at the 0.01 significance level, while soy had the least.

	Crop	Scenario	Countries	Significance
0	Wheat	base	Pakistan, Ukraine, Syria, Jordan, Cyprus, Mongolia, Russia, Belarus, Armenia, Israel, Azerbaijan, Somalia, Türkiye, Seychelles, Kyrgyz Republic, Libya, Lebanon, Egypt, Georgia, Moldova, Romania	0.166471
1	Wheat	ASRS	Pakistan, Ukraine, Syria, Jordan, Cyprus, Mongolia, Russia, Belarus, Armenia, Israel, Azerbaijan, Somalia, Türkiye, Seychelles, Kyrgyz Republic, Libya, Lebanon, Egypt, Georgia, Moldova, Romania	0.166364
2	Rice	base	India, Iran, Burkina Faso, Jordan, Kuwait, Benin, Senegal, Saudi Arabia, Sri Lanka, Seychelles, Madagascar, Cameroon, Nigeria, Maldives, Togo, Qatar, Bahrain, South Sudan, Bangladesh, Yemen, Comoros, Nepal, Central African Republic, Congo Republic, Guinea, Mali, Eritrea, Somalia, Ethiopia, Angola, Djibouti, Liberia, Sudan, Algeria, Bhutan	0.175240
3	Rice	base	Marshall Islands, St. Kitts and Nevis, Dominican Republic, Zimbabwe, Iceland, Morocco, South Africa, Japan, Botswana, Colombia, Thailand, Samoa, Vanuatu, Israel, Ecuador, Namibia, Mozambique, New Caledonia, New Zealand, Panama, Lesotho, Equatorial Guinea, Australia, Iraq, Bahamas, Belize, Nicaragua, Tunisia, Canada, United States, Honduras, Guatemala, Haiti, Mauritania, Kiribati, Laos, Hong Kong, Gabon, Taiwan, Eswatini	0.175240
4	Rice	base	Cote d'Ivoire, Malaysia, Singapore, Vietnam, Fiji, Macau, Nauru, French Polynesia, Philippines, Indonesia	0.175240
5	Rice	ASRS	India, Iran, Burkina Faso, Kuwait, Benin, Senegal, Russia, Saudi Arabia, Sri Lanka, Armenia, Seychelles, Madagascar, Cameroon, Nigeria, Maldives, Togo, Qatar, Bahrain, South Sudan, Bangladesh, Yemen, Comoros, Nepal, Central African Republic, Congo Republic, Guinea, Mali, Eritrea, Somalia, Ethiopia, Angola, Djibouti, Liberia, Sudan, Algeria, Bhutan	0.214687
6	Rice	ASRS	Marshall Islands, Gabon, Dominican Republic, Zimbabwe, Iceland, South Africa, Morocco, Botswana, Japan, Colombia, Samoa, Ecuador, Vanuatu, Thailand, Israel, Namibia, Mozambique, New Caledonia, New Zealand, Lesotho, Equatorial Guinea, Australia, Iraq, Bahamas, Belize, Nicaragua, Tunisia, Canada, United States, Honduras, Hong Kong, Guatemala, Mauritania, Kiribati, Norway, Laos, Panama, Haiti, Taiwan, Eswatini	0.121229
7	Rice	ASRS	Cote d'Ivoire, Malaysia, Singapore, Vietnam, Fiji, Macau, Nauru, French Polynesia, Philippines, Indonesia	0.181939
8	Soya beans	base	Pakistan, Cabo Verde, Iran, Brazil, China, Jordan, Bangladesh, United Kingdom, Yemen, Mongolia, Portugal, North Korea, Iceland, Russia, Congo Republic, Saudi Arabia, Thailand, Israel, Vietnam, Netherlands, Norway, Greece, Malta, Spain	0.015533
9	Soya beans	base	Singapore, Austria, Tanzania, Macau, Gabon, Zimbabwe, Croatia, Tajikistan, Zambia, Botswana, Belarus, Armenia, Switzerland, Azerbaijan, Sri Lanka, Mozambique, Türkiye, Sweden, Serbia, Italy, Ireland, Cote d'Ivoire, Syria, Qatar, Hungary, Turkmenistan, Slovenia, Mali, Bulgaria, United Arab Emirates, Lebanon, Ethiopia, Uganda, Hong Kong, Poland, Algeria, India, Burkina Faso, Lithuania, Luxembourg, Suriname, Belgium, Kuwait, Senegal, South Africa, Kazakhstan, Ghana, Slovakia, Namibia, Rwanda, Uzbekistan, Finland, Montenegro, New Zealand, Kyrgyz Republic, Malawi, Nigeria, Georgia, Moldova, Lesotho, DR Congo, Romania, Equatorial Guinea, Ukraine, Bosnia and Herzegovina, Togo, Latvia, Cyprus, Canada, Bahrain, France, Mauritius, Estonia, Afghanistan, Angola, Liberia, Kenya, Denmark, Brunei Darussalam	0.131546
10	Soya beans	ASRS	Pakistan, Cabo Verde, Iran, Brazil, China, Jordan, Belgium, Bangladesh, United Kingdom, Mongolia, Portugal, North Korea, Iceland, Russia, Congo Republic, Saudi Arabia, Thailand, Israel, Vietnam, Netherlands, Norway, Greece, Malta, Gabon, Algeria, Spain	0.079711
11	Soya beans	ASRS	Austria, Tanzania, Zimbabwe, Croatia, Tajikistan, Zambia, Botswana, Belarus, Armenia, Switzerland, Azerbaijan, Sri Lanka, Mozambique, Türkiye, Sweden, Serbia, Italy, Cote d'Ivoire, Syria, Qatar, Hungary, Turkmenistan, Slovenia, Mali, Bulgaria, United Arab Emirates, Lebanon, Ethiopia, Uganda, Poland, India, Burkina Faso, Lithuania, Luxembourg, Kuwait, Senegal, Kazakhstan, South Africa, Slovakia, Namibia, Rwanda, Uzbekistan, Finland, Montenegro, Kyrgyz Republic, Malawi, Nigeria, Georgia, Moldova, Lesotho, DR Congo, Romania, Equatorial Guinea, Bosnia and Herzegovina, Ukraine, Togo, Latvia, Cyprus, Bahrain, France, Yemen, Estonia, Afghanistan, Angola, Liberia, Kenya, Denmark, Ghana	0.029174
12	Soya beans	ASRS	Malaysia, Singapore, South Korea, St. Vincent and the Grenadines, Costa Rica, St. Kitts and Nevis, Suriname, Macau, Dominican Republic, Philippines, Germany, Papua New Guinea, Mexico, Morocco, Japan, Trinidad and Tobago, Colombia, Ecuador, New Caledonia, New Zealand, Madagascar, Venezuela, Egypt, Cambodia, El Salvador, Australia, Ireland, Myanmar, Iraq, Belize, Nicaragua, Tunisia, Jamaica, Canada, United States, Grenada, Mauritius, St. Lucia, Honduras, Hong Kong, Nepal, Guatemala, Indonesia, Czechia, Guyana, Cuba, Laos, Timor-Leste, Panama, Barbados, Brunei Darussalam, Taiwan	0.135465

Figure 12: The members of all 5 Tg insignificant communities, including their respective crop and scenario.

	Crop	Scenario	Countries	Significance
0	Wheat	base	Türkiye, Georgia, Pakistan, Cyprus, Lebanon, Jordan, Armenia, Libya, Israel, Belarus, Egypt, Seychelles, Russia, Kyrgyz Republic, Azerbaijan, Somalia, Moldova, Romania, Syria, Ukraine, Mongolia	0.166471
1	Rice	base	Guinea, Djibouti, Jordan, South Sudan, Eritrea, Bangladesh, Maldives, Algeria, Ethiopia, Congo Republic, India, Seychelles, Benin, Madagascar, Iran, Bhutan, Yemen, Nepal, Somalia, Saudi Arabia, Liberia, Angola, Burkina Faso, Cameroon, Qatar, Comoros, Bahrain, Central African Republic, Togo, Senegal, Nigeria, Mali, Sudan, Sri Lanka, Kuwait	0.175240
2	Rice	base	Gabon, New Zealand, Canada, Iraq, Honduras, Vanuatu, Dominican Republic, St. Kitts and Nevis, Mozambique, Taiwan, Marshall Islands, Zimbabwe, Israel, Nicaragua, Morocco, United States, Tunisia, Bahamas, Mauritania, Panama, Iceland, Laos, Hong Kong, Guatemala, Lesotho, Botswana, South Africa, Namibia, Ecuador, Thailand, Samoa, Colombia, Japan, Equatorial Guinea, Australia, Haiti, Belize, New Caledonia, Kiribati, Eswatini	0.175240
3	Rice	base	Fiji, Indonesia, Singapore, Cote d'Ivoire, French Polynesia, Malaysia, Nauru, Philippines, Macau, Vietnam	0.175240
4	Rice	ASRS	Guinea, Cote d'Ivoire, Djibouti, South Sudan, Eritrea, Mozambique, Bangladesh, Maldives, Algeria, DR Congo, Ghana, Ethiopia, Congo Republic, India, Seychelles, Benin, Madagascar, Iran, Bhutan, Yemen, Nepal, Somalia, Timor-Leste, Saudi Arabia, Liberia, Angola, Burkina Faso, Cameroon, Qatar, Comoros, Bahrain, Egypt, Russia, Central African Republic, Togo, Senegal, Oman, Nigeria, Mali, Sierra Leone, Sudan, Sri Lanka, Kuwait	0.080868
5	Soya beans	base	Cabo Verde, Saudi Arabia, Jordan, Pakistan, Norway, Greece, United Kingdom, Bangladesh, Brazil, Israel, Spain, Malta, China, Thailand, Congo Republic, Russia, North Korea, Iceland, Portugal, Iran, Yemen, Mongolia, Netherlands, Vietnam	0.015533
6	Soya beans	base	Canada, Hungary, Zimbabwe, Algeria, Ghana, Kyrgyz Republic, Moldova, Slovakia, Ukraine, Lesotho, Kenya, Burkina Faso, Angola, Armenia, South Africa, Namibia, Croatia, Turkmenistan, Rwanda, Austria, Togo, Finland, Nigeria, Romania, Ireland, Malawi, Sri Lanka, Kuwait, Serbia, Türkiye, Kazakhstan, Gabon, New Zealand, Cote d'Ivoire, Georgia, Lithuania, Estonia, France, Mozambique, Uzbekistan, Switzerland, Sweden, Belarus, DR Congo, Italy, India, Ethiopia, Brunei Darussalam, Tanzania, Singapore, Afghanistan, Azerbaijan, Zambia, Hong Kong, Tajikistan, Suriname, Bulgaria, Liberia, Montenegro, Cyprus, Denmark, Lebanon, Botswana, Qatar, Mauritius, Latvia, Macau, Bosnia and Herzegovina, Luxembourg, Bahrain, Poland, Senegal, United Arab Emirates, Uganda, Belgium, Equatorial Guinea, Slovenia, Mali, Syria	0.131546
7	Soya beans	ASRS	Türkiye, Gabon, Jordan, Bangladesh, Israel, Sweden, Algeria, Malta, Spain, Tunisia, Congo Republic, Iceland, Portugal, Iran, Guyana, Yemen, Netherlands, Cabo Verde, Saudi Arabia, Cyprus, Pakistan, Lebanon, Norway, Greece, United Kingdom, Brazil, Thailand, China, Russia, North Korea, South Korea, Belgium, Syria, Mongolia, Vietnam	0.018015
8	Maize	ASRS	New Zealand, Canada, Barbados, French Polynesia, Honduras, Norway, St. Lucia, St. Kitts and Nevis, United States, China, Bahamas, Antigua and Barbuda, Fiji, Laos, Guyana, Jamaica, St. Vincent and the Grenadines, Mexico, Trinidad and Tobago, Grenada, Dominica, Belize, Mongolia	1.000000

Figure 13: The members of all 47 Tg insignificant communities, including their respective crop and scenario.

	Vulnerability Score	Standardized Vulnerability
Sweden	749.080691	4.733622
Finland	612.600008	3.730002
Norway	562.137732	3.358925
Mongolia	476.037967	2.725784
Canada	468.369616	2.669395
Georgia	355.385467	1.838558
Switzerland	355.045824	1.836061
Iceland	350.645966	1.803706
United Kingdom	350.292542	1.801107
Oman	339.369424	1.720783
United States	337.208843	1.704895
Kazakhstan	331.890933	1.665790
Russia	331.227660	1.660912
Azerbaijan	321.932589	1.592560
Gambia	320.220177	1.579968
North Korea	317.582812	1.560574
Hong Kong	309.422925	1.500570
Tajikistan	282.401170	1.301863
Syria	279.098161	1.277574
Spain	277.848361	1.268384

Figure 14: Country scores for $V_v(v)$ in the 5 Tg scenario, ranked from greatest to least and standardized. The top five vulnerable states post-disaster are Sweden, Finland, Norway, Mongolia, and Canada.

	Vulnerability Score	Standardized Vulnerability
Ukraine	2192.169695	2.220857
Switzerland	2078.099903	2.010198
Tajikistan	2066.494310	1.988766
Sweden	2056.975273	1.971186
Kazakhstan	2041.989038	1.943511
Azerbaijan	1996.333365	1.859196
Japan	1996.319083	1.859170
Canada	1963.440020	1.798450
Finland	1959.731881	1.791602
Georgia	1957.271103	1.787058
Uzbekistan	1955.798852	1.784339
Latvia	1952.142371	1.777586
Armenia	1946.542495	1.767245
Turkmenistan	1938.761485	1.752875
France	1932.193070	1.740745
Mongolia	1837.376520	1.565643
Germany	1812.722378	1.520113
Mexico	1802.487251	1.501211
Russia	1795.849143	1.488952
United States	1790.301245	1.478707

Figure 15: Country scores for $V_v(v)$ in the 47 Tg scenario, ranked from greatest to least and standardized. The top five vulnerable states post-disaster are Ukraine, Switzerland, Tajikistan, Sweden, and Kazakhstan. There scores often more than double that of the 5 Tg weight.

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